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Components of an environment – biotic and abiotic

The biotic components of an environment are the organisms present.

1 What are the *abiotic* components of an environment?

For scientific purposes, the living world has been divided into groups called *biomes*, meaning 'zone of life'. Biomes are large, naturally occurring communities of plants and animals forming recognisable groups occupying a large territory. The two major world biomes are aquatic and terrestrial. They can be divided into more specific categories based on the species present.

Aquatic (*aqua*, water): These are the 'wet' biomes and consist of freshwater (eg ponds, rivers) and marine (eg sea/ocean, estuary). The composition of the water, which differs between each, affects the kinds of species that will be found.

2 What is the main difference between freshwater and marine conditions?

Desert (*desertare*, to forsake): Roughly 20% of the Earth is desert. A biome that commonly receives less than 50 cm of precipitation per year is classified as a desert. This lack of water means that many types of vegetation cannot survive and deserts are recognisable by the vast stretches of sand on which little or no vegetation is present. Any vegetation present has evolved to adapt to the arid conditions.

Forest (*forestis*, wood): Forests are biomes that are densely covered by various tree species (depending on their geographical location). Smaller trees and shrubs are usually found beneath the tree canopy. Forests cover about 30% of the Earth and represent 70% of the carbon found in living organisms.

Tundra (*tanturia*, treeless plain): Commonly found in cold northern regions of North America, parts of Asia and the Arctic districts of Europe, tundra are large, treeless areas. At ground level, the soil is cold and hard. As expected, these are the most frigid of the biomes. *Arctic tundra* is found in the Northern Hemisphere (eg North Pole) and may be flat or slightly rolling – but not mountainous, while *alpine tundra* is that which has an alpine, or mountainous, landscape. The combination of high altitudes and cold temperatures makes it impossible for trees to grow on alpine tundra.

Grassland: As the name suggests, grasslands are large areas covered by grasses.

Savanna: Savannas are often mistaken for grasslands but they can be distinguished by the presence of an occasional tree or shrub. More than half of Africa's landscape is savanna. The average rainfall is 50–130 cm per year, most of which comes in a fairly short space of time followed by long periods of hot, dry conditions.

3 The North Pole and Antarctica have often been compared to desert. Suggest a reason why.

4 Give a definition for *arid*.

5 Give a definition for *canopy*.

6 Suggest a meaning for *graminivorous* then check with your dictionary to see if you were correct.

Give definitions for the following biomes, including naming an area where each can be found:

7 mangrove

8 reef

9 rainforest

Research

10 Investigate and describe what *spinifex* and *tussock* are.

11 Another type of biome is the *sclerophyll forest*. Briefly describe the differences between wet sclerophyll and dry sclerophyll (do not simply state that one is wet, the other dry – describe rainfall and vegetation).

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